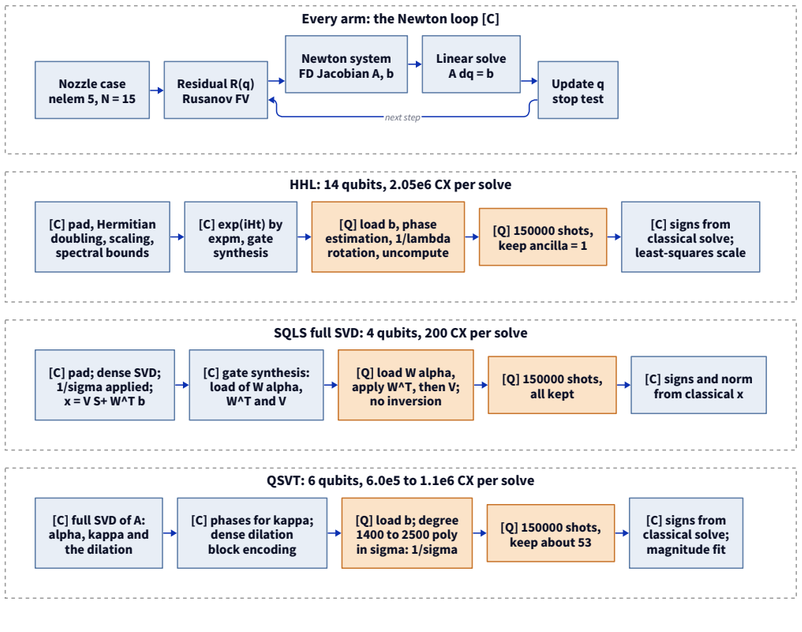
SQLS, HHL and QSVT as the Newton linear solver of the quasi-1D FVM Euler nozzle

nelem 5, 150000 shots, seeds 1, 2, 3, 6, 7 – internal, 2026-11-04

Question. On one code line (fvm_euler_1d_solver `d4a531b`, QSVT at `219907c`, which contains it; qiskit 2.6.0), how do SQLS, HHL and QSVT compare as the linear solver of each Newton step on the November 2025 case: HHL at its November defaults and at its best shots settings, SQLS and QSVT at their best?



The Newton loop is classical and shared; only the linear solve differs. Classical steps blue-grey, quantum-circuit steps orange. The circuits run on the Aer statevector simulator.

What each pathway does, and where

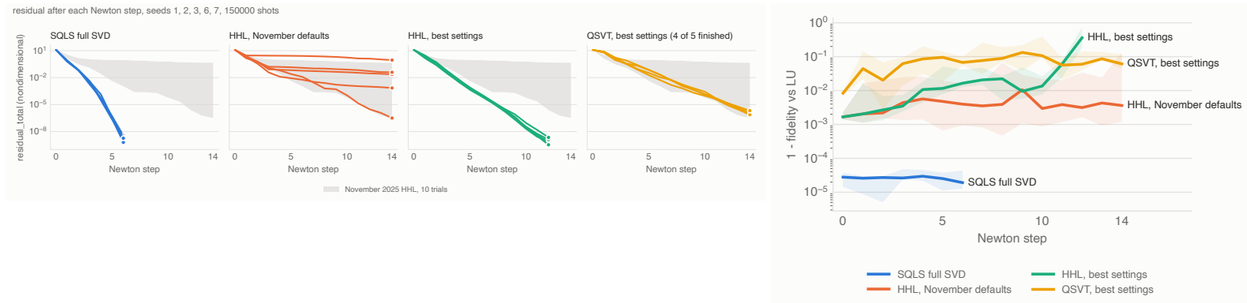
C: classical, on the host. Q: on the circuit (simulated).

Stage	HHL, November defaults	HHL, best settings	SQLS, full SVD	QSVT, best settings
Pre-processing	C: pad to 16 (b pad 10^{-8}); Hermitian doubling to 32 x 32; spectral scaling; κ, eigenvalue bounds	same, b pad 0	C: pad to 16 (b pad 0); dense SVD $A = WSV^T$; $x = VS^+W^Tb$ known	C: pad to 16 (b pad 0); full SVD of A gives α, κ and the dilation; phases for κ
Gate synthesis	C: e^{iHt} by matrix exponential; controlled powers to gates	same	C: load of $W\alpha$; dense W^T, V	C: dense dilation block encoding of $A^\dagger/\alpha$
On the circuit	Q: load b; phase estimation (8 clock qubits); $1/\lambda$ rotation; uncompute	same	Q: load $W\alpha$; apply W^T, V	Q: load b; alternate the block encoding and phase rotations, degree 1400 to 2500
Inversion	Q $(1/\lambda)$	Q	C $(1/\sigma)$	Q (polynomial $\approx 1/\sigma$)
Measurement	Q: 150000 shots, keep ancilla = 1	same	Q: 150000 shots, all kept	Q: 150000 shots, keep about 53 (postselected)
Signs	C: classical Hermitian solve, after the fit	C: same, before the fit	C: from the classical x	C: classical solve, before the fit
Scale	C: least squares on A, b	same	C: $\|x\|$ from the SVD	C: magnitude fit
Per solve	14 qubits (13 on 3 % of solves), 2.05×10^6 CX; circuit time 252 s (1 thread)	same; 248 s (2 threads)	4 qubits, 200 CX; 0.23 s	6 qubits, 6.0×10^5 to 1.1×10^6 CX; 0.63 s

Results

Median (range) over the 5 seeds (4 finished for QSVT); residual_total is nondimensional; the update ratio is against the LU solve of A; fidelity is against an LU solve of the same system (for HHL, of its Hermitian form).

	SQLS full SVD	HHL, November defaults	HHL, best settings	QSVT, best settings	HHL, November 2025 (10 trials, older code, hpc-cluster)
Newton solves	7 (converged)	15 (cap)	13 (converged)	15 (cap); seed 3 failed at step 1	15 (cap)
residual after step 4	5.3×10^{-5} (2.9×10^{-5} to 1.4×10^{-4})	0.075 (0.013 to 2.8)	0.0049 (0.0033 to 0.0062)	0.15 (0.076 to 0.25)	0.34 (0.026 to 0.98)
final residual (step 14, or at the stop)	1.8×10^{-9} (6.3×10^{-10} to 3.5×10^{-9})	0.023 (3.2×10^{-7} to 0.90)	9.5×10^{-10} (3.3×10^{-10} to 2.2×10^{-9})	1.0×10^{-6} (7.0×10^{-7} to 2.1×10^{-6})	0.0067 (3.2×10^{-7} to 0.46)
median $\|dq\| / \|dq_{LU}\|$	1.00	0.13 (0.072 to 0.78)	0.95 (0.94 to 0.98)	0.91 (0.89 to 1.01)	0.22 (0.055 to 0.98)
stalled runs (ratio below 0.1)	0 of 5	1 of 5	0 of 5	0 of 4	2 of 10
median fidelity (median over runs)	0.99997	0.996	0.988	0.928	0.992
lowest fidelity while residual $> 10^{-6}$	0.99995	0.933	0.949	0.605	0.927
case wall time per run	6.1 s	5323 to 6051 s	3708 to 3862 s	279 to 291 s	about 5900 s circuit time



Left: residual after each Newton step, one line per seed; grey band: the November 2025 HHL trials. Right: infidelity against LU per Newton step, median over seeds, band from lowest to highest.

Reading

- **SQLS converges as LU does**: 7 Newton steps to about 10^{-9}, every update the LU size. Its classical pre-processing already holds the solution, and its readout takes the signs and the norm from it; the circuit adds only sampling noise on the magnitudes ($1 - F \approx 2.6 \times 10^{-5}$ at 150000 shots).
- **HHL at the November defaults stalls on the update's size, not its direction**: fidelity 0.996, but the update is a median 13 % of the LU step, since the scale is fitted before the signs are applied. One seed in five stalls, as in November.
- **HHL at its best settings converges every seed** in 13 steps to about 10^{-9}, the stalled seed 7 among them: signs applied before the fit and a zero power-of-2 pad bring the update to 94 % to 98 % of the LU step. Its fidelity drops only on the last one or two solves, below the stop threshold.
- **QSVT inverts on the circuit with 6 qubits and no phase estimation, but at 150000 shots it is shot-starved**: postselection keeps a median 53 shots per solve, so its updates have the right size (0.91 of LU) but a noisy direction (median fidelity 0.928). It descends steadily to about 10^{-6} by step 14 (it needs about 18 steps here), and seed 3's first update drove a cell's pressure negative and broke the run. In statevector mode its error against LU is 60 times smaller than HHL's. Tested at more shots (seed 1): 11 Newton steps at 1.6×10^6 shots, 9 at 7.5×10^6, as the shot-budget model predicted.
- **Both lean on classical computation**: HHL's spectral scaling, eigenvalue bounds, matrix exponentials for its circuit blocks and a Hermitian LU for the signs are all classical; its inversion $(1/\lambda)$ is on the circuit. SQLS inverts $(1/\sigma)$ classically and copies signs and norm from that exact solution. The 10^4 gap in CX compares an inversion with a state load.
- **Limits**: the readouts are not matched (HHL keeps a least-squares scale, SQLS the exact norm); 5 seeds per arm, one shot count, nelem 5, ideal simulator; the two HHL switches changed together.

Provenance. fvm_euler_1d_solver `d4a531bd` (feature/sqls-aiaa-4q26), sqls `67057e4b`, quantum_linear_solvers `99231b85`, qiskit 2.6.0, qiskit-aer 0.17.2; run on linux-server, 2026-11-03 and 2026-11-04. November 2025: fvm `a5317f2b`, qiskit 1.2.4, on an hpc-cluster node, unseeded. QSVT: fvm `219907cf` (feature/sqls-aiaa-4q26-svd-deflation), change add-qsvt-arm-nelem5-shots. Change compare-hhl-sqls-nelem5-shots in proj:aiaa-4q26.